AI Programming With Python Assignment 01

Submission instructions :

1. First of all, watch the assignment instruction video very carefully
2. Create a Google Collaboratory File, write all of the answers of the questions in that single .ipynb (colab) file
3. For each question create a **Text cell** with the question number and then a **Code cell** containing the solution.
4. Print or return sample outputs shown in the question so graders can verify results easily.
5. Share the colab file in '**Anyone with the link**' & '**Editor**' Mode , copy the link and just submit that link

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Question : 1	**20 Marks**

Given a list of integers, extract all numbers that are divisible by 4 but not divisible by 8.

Input Format:
Space-separated integers

Output Format:
A list of numbers satisfying the condition

Example:
Input:
```
4 6 8 12 16 20 24 28
```

Output:
```
[4, 12, 20, 28]
```

Question : 2	**20 Marks**

Write a function `compute_ratio(x, y)` that returns `(x + y) / (x * y)`. Use exception handling to manage division by zero.

Use exception handling to manage:

- Division by zero

Input Format:
Two values

Output Format:
Computed result OR error message

Example:
Input:
`5 0`

Output:
Error: Division not possible

Question : 3	20 Marks

Write a Python program that reads a text file and finds the longest word.

Requirements:

- Open and read data from a file (Example: `data.txt`)
- If multiple words have the same longest length, print any one of them

Input Format:
A text file containing multiple lines of text

Output Format:
The longest word

Example File Content (data.txt):
```
Python is a wonderful language for machine learning.
```

Output:
```
wonderful
```

Question : 4	20 Marks

Write a program that converts a sentence into a list of words but only includes words that start with a vowel (a, e, i, o, u), ignoring case.

Input Format:

A single line sentence

OutputFormat:
A list of strings

Example:
Input:
```
An apple a day keeps the doctor away
```
Output:
```
['An', 'apple', 'a', 'away']
```

Question : 5	20 Marks

A word is called a **Palindrome** if reversing its letters results in the same word (ignoring case). Write a program to check whether a given string is a palindrome.

Input Format:
A single word string

Output Format:
`"Palindrome"` or `"Not Palindrome"`

Example:
Input:
```
Radar
```

Output:
```
Palindrome
```